

AgiBot X2 — Progress Update

Box-Pickup Motion-Retargeting Policy | On-Robot VLM Perception

Whole-body manipulation via human MoCap retargeting + vision-language perception benchmark

Box-Pickup Policy — Objective & Approach

- Task: whole-body box pickup — walk up, two-hand grasp, lift, carry, and set down a large box.
- Reference source: human motion-capture clip (OmniRetarget dataset, sub3_largebox_003, SMPL-H 52-joint model).
- Embodiment switch: motion authored on a human SMPL-H skeleton is retargeted onto the AgiBot X2 — 31 DOF (6+6 legs, 3 waist, 7+7 arms, 2 head), 1.4 m tall, rigid half-sphere palms (no fingers).
- Status: policy is currently in training; deployment on the physical X2 is planned once this run completes.

Pipeline



Blind whole-body policy (no cameras); object state used only by the critic during training.

Retargeting Logic

- Method: interaction-mesh retargeting solved as a per-frame convex QP with SQP linearization (CVXPY + Clarabel).
- 14-landmark SMPL-H → X2 correspondence (pelvis, hips, knees, ankles, toes, shoulders, elbows, wrists → hand-contact links).
- Synthetic hand-contact link at (0.02, 0, -0.13) m from each wrist so the human wrist maps to the grasp surface.
- Constraints: Laplacian (interaction-mesh) preservation; non-penetration (floor + box); foot-stick / foot-lock; joint limits; trust region; self-collision.

Per-frame QP objective

$$\min_{\Delta q} \|L(q) - L_{human}\|^2 + w_s \|\Delta q\|^2 \quad s.t. \quad \|\Delta q\| \leq \delta, \quad q^- \leq q \leq q^+$$

$L(\cdot)$: interaction-mesh Laplacian · δ : trust-region step · $q_{\pm}$: joint limits

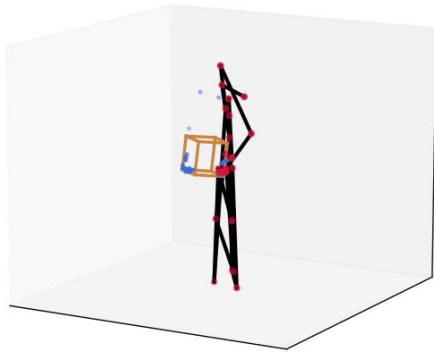
Figure placeholder

Add interaction-mesh / retargeting figure from the paper

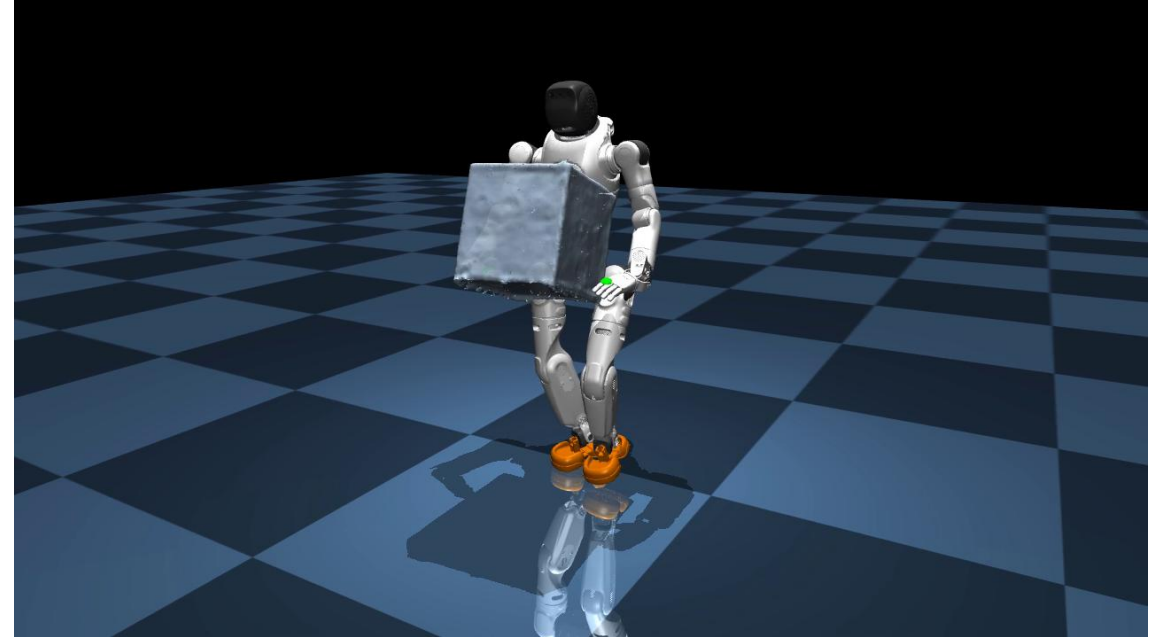
Reference Motion: Human MoCap → Retargeted X2

Source — human SMPL-H MoCap

OMOMO human mocap: sub3_largebox_003 (frame 98/196, 30 fps)



Retargeted — AgiBot X2 (31-DOF)



Same clip before and after the interaction-mesh IK pipeline; contact states corrected post-retargeting. (Click to play.)

Policy Training — RL Tracking

4096

parallel envs

50 Hz

control (200 Hz sim)

31

actuated DOF

PPO

 $\lambda=0.95 \cdot \gamma=0.99$

- Framework: IsaacLab (Isaac Sim) + holosoma.
- Actor: blind MLP 512-256-128 (ELU) — prev action, base ang-vel, dof pos/vel, motion phase, ref orientation.
- Critic: privileged obs adds base lin-vel + object & body state.
- Key reward: two-hand-to-box proximity as a product term — both palms must reach the box simultaneously.
- Status: currently training; deploy on physical X2 after the run.

Actor (blind) → 512-256-128 → 31 DOF

Critic (privileged: + object & body state)

Two-hand grasp reward

$$r_{hands} = \exp\left(-\frac{\sum_{i \in \{L, R\}} \|p_i^{hand} - p^{box}\|^2}{\sigma^2}\right), \quad \sigma = 0.25 \text{ m}$$

Product (not mean) over hands forces bimanual contact.

VLM Perception — Setup

- Goal: scene understanding from the X2 head RGB camera; frames streamed to a workstation for evaluation.
- Three VLMs served locally via Ollama: Qwen2.5-VL 7B, Qwen2.5-VL 3B, Moondream (~1.8B).
- Protocol: 15 s clip @ ~20 fps → 12 evenly-spaced frames; identical prompt, temperature 0.2; one model resident at a time (8 GB VRAM).

Prompt (identical for all models)

"You are a robot's vision system. List the main objects and people you see and roughly where they are (left/center/right). Then one short summary sentence. Be concise."

VLM Performance Metrics

Model	Params	Mean lat.	Median	Rate	Gen speed
Qwen2.5-VL 7B	7 B	2.87 s	2.89 s	0.35 Hz	51.3 tok/s
Qwen2.5-VL 3B	3 B	2.45 s	2.50 s	0.41 Hz	107.6 tok/s
Moondream	~1.8 B	0.61 s	0.62 s	1.65 Hz	215.4 tok/s

- Warm, end-to-end latencies on an RTX 5070 (8 GB); 12 frames, single 15 s clip.
- Moondream is ~4-5× faster — the only model near real-time (~1.7 Hz).
- Qwen2.5-VL 7B gives the most accurate scene descriptions; 3B is slower than Moondream yet less accurate on this scene.

VLM Qualitative Comparison — Frame 224 (t = 11.3 s)



Qwen2.5-VL 7B (2.80 s | 51.6 tok/s)

Objects: yellow toolbox, green cord, wheels, white t-shirt with text, gray pants. People: man in white t-shirt squatting. Summary: a person interacting with a wheeled device in an industrial setting.

Qwen2.5-VL 3B (2.60 s | 107.9 tok/s)

A person squatting in the center of the room; yellow and black robotic feet on the floor, likely attached to a robot or machine.

Moondream (0.67 s | 203.4 tok/s)

Person in white shirt, gray pants, black shoes, and yellow shoes standing near a desk with a green cord on it.